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To the Editor-in-Chief and Associate Editor
IEEE Transactions on Information Theory

Dear Editor:

Please consider the enclosed manuscript, "Tradeoffs Between Rate and Conditional Content with Encoder-Observed Context," for publication in the IEEE Transactions on Information Theory. I am the sole author.

The paper evaluates Steinberg's common-reconstruction functional on an augmented Gaussian source: the encoder observes a pair (Y, V), the distortion binds Y alone, and a third party holds a noisy copy of the context V that no decoder uses. Two results are primary: a closed form for the minimal conditional content, the larger root of an explicit quadratic, for Y and V scalar linear functionals of an arbitrary-dimensional jointly Gaussian source; and the exact region of jointly achievable rate and conditional content, with a two-water-level Pareto frontier and a strict tradeoff exactly when the described variable and the context are partially correlated, persisting at the clean-context boundary. A structural consequence is non-determination: the function is not a functional of the reduced source–side-information marginal, so the encoder's access to the context changes the operational problem. Extensions give the frontier with vector context, the binary rate–content frontier through a tilt equation, and a direct Gaussian operational theorem whose achievability quantizes the conditioner, the source, and the reproduction and invokes the finite-alphabet theorem. Consistency anchors recover the classical, Gray, and Steinberg corners.

Two earlier documents are relevant, and they are distinct from one another. First, this Transactions declined a different manuscript of mine, IT-26-0996, on 27 July 2026 without review, on scope and presentation; it was a broad synthesis. I ask the editors to read the present submission independently of it: this is a single-object work in the classical rate–distortion genre and shares no theorem with that declined submission. The terminology concerns raised in that review have been addressed here by adopting standard source-coding vocabulary throughout.

Second, and separately, the manuscript cites an archived preliminary manuscript by the author (Zenodo record, doi:10.5281/zenodo.21776291), as disclosed in its Related Work section. The present manuscript's own sources, verification scripts, and reproduction notebook are archived separately (doi:10.5281/zenodo.22678341). The discrete operational region theorem and its single-letterization appeared there, as did the scalar Gaussian corner, and both are inherited here with attribution. The material from Section III onward is developed here; two components have close predecessors that are credited in place, namely the pair-sufficiency reduction (a distortion-preserving specialization of Armstrong's information-bottleneck sufficiency) and the binary conditional-content function (which coincides with the common-reconstruction rate of Lu et al., the contribution there being the joint rate–content frontier). The present paper supersedes the source-coding content of that record, which remains the archived evidence artifact.

All closed forms are verified by an apparatus available as supplementary repository material: symbolic and numerical verification scripts, including two written independently of the derivations, a partial Lean 4 formalization of the core algebra, and MATLAB symbolic checks. A prior-art screen conducted in September 2026 is documented in the repository.

This manuscript is not under review elsewhere, and I have no conflicts of interest to declare.

Sincerely,

Andrew H. Bond
Senior Member, IEEE